AI-Based Water Quality Monitoring and Pool Maintenance System

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Project Proposal Report

N M Shafan - IT22248480

Swimmer Behavior and Drowning Detection Module

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
N M Shafan	IT22248480	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof. Anuradha Karunasena	13/03/2026	

ABSTRACT

Swimming pools require continuous monitoring and maintenance to ensure safe water quality and hygienic conditions for users. Poorly maintained pools can lead to health risks such as bacterial infections, chemical imbalances, and water contamination. In addition to water quality concerns, swimmer safety is also a critical factor that requires constant supervision. Many existing pool management systems rely on manual monitoring and fixed maintenance schedules, which may not accurately reflect real-time water conditions, swimmer behavior, or pool usage. These limitations often result in inefficient maintenance practices, excessive chemical usage, delayed responses to unsafe water conditions, and difficulty in identifying emergency situations such as drowning.

This research proposes an **AI-Based Water Quality Monitoring and Pool Maintenance System** that integrates sensor-based monitoring, intelligent data analysis, and computer vision technologies to improve the efficiency and safety of pool management operations. The proposed system collects real-time water quality data, including parameters such as pH level, turbidity, chlorine concentration, and temperature, using IoT-based sensors. Additionally, the system analyzes pool usage patterns and operational data to better understand how swimmer activity affects water quality. As part of the system, a computer vision-based module is used to monitor swimmer behavior and detect potential drowning situations through camera-based observation.

The collected data will be processed using machine learning, computer vision, and data analysis techniques to identify patterns and predict potential water quality deterioration as well as abnormal swimmer behaviors. Based on these predictions, the system will generate maintenance alerts, safety notifications, and recommendations for optimal pool cleaning schedules and chemical treatments. This predictive and monitoring approach helps pool managers maintain safe water conditions while improving swimmer safety and minimizing unnecessary maintenance activities.

The expected outcome of this research is the development of an intelligent monitoring system that improves pool safety, reduces operational costs, and enhances maintenance efficiency. The system also contributes to the advancement of smart facility management solutions by integrating real-time monitoring, predictive analytics, automated decision support, and AI-based swimmer behavior analysis.

Keywords : Water Quality Monitoring, Pool Maintenance, Artificial Intelligence, Internet of Things (IoT), Computer Vision, Drowning Detection, Predictive Maintenance

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LIST OF ABBREVIATIONS

Abbreviation	**Meaning**
AI	Artificial Intelligence
IoT	Internet of Things
ML	Machine Learning
pH	Potential of Hydrogen
WQMS	Water Quality Monitoring System
RP	Research Project
NSF	Novel System Feature

INTRODUCTION

1.1 Background and Context

Swimming pools are commonly found in recreational centers, hotels, sports complexes, and educational institutions, where they provide opportunities for leisure, exercise, and training. Ensuring a safe swimming environment is an important responsibility for facility operators. While maintaining appropriate water quality is essential for hygiene, monitoring the safety of swimmers is equally important to prevent accidents and protect the well-being of pool users.

In most swimming pool facilities, swimmer safety is mainly supervised by lifeguards who observe the activities of people in the pool. Although lifeguards are trained to identify risky situations, continuous monitoring of all swimmers can be difficult, especially in crowded environments. Human observation alone may not always be sufficient, as distractions, fatigue, or limited visibility can delay the identification of swimmers who are experiencing distress. Such delays may increase the risk of serious incidents such as drowning.

With the advancement of modern technologies, automated monitoring systems have been introduced to support safety management in various environments. Computer vision technology, in particular, enables machines to interpret visual information from cameras and analyze human behavior. In swimming pools, camera-based systems can capture continuous video footage of the pool area, which can then be analyzed to detect movement patterns and behavioral changes among swimmers. However, many existing surveillance systems primarily focus on general security monitoring and lack intelligent mechanisms to analyze swimmer behavior or identify potential drowning situations in real time.

To overcome these limitations, this research is carried out as part of the **AI-Based Water Quality Monitoring and Pool Maintenance System**, which aims to enhance pool management through the integration of intelligent monitoring technologies. Within this overall system, the **Computer Vision-Based Swimmer Behavior and Drowning Detection component** focuses on improving swimmer safety by automatically observing activities in the pool and identifying unusual behavioral patterns.
The proposed component utilizes video data captured from surveillance cameras to analyze swimmer movements and body positions. By applying computer vision and machine learning techniques, the system can distinguish between normal swimming activities and behaviors that may indicate distress or potential drowning. When abnormal patterns are detected, the system can generate timely alerts to notify lifeguards or pool operators, enabling quicker intervention and reducing the likelihood of severe accidents.

In addition, this research supports **Sustainable Development Goal (SDG) 3: Good Health and Well-being**, which emphasizes the importance of protecting human health and promoting safe environments. The study also contributes to the **Centre of Excellence in Artificial Intelligence (CEAI)** research cluster by applying intelligent visual analysis techniques to enhance safety monitoring and smart facility management within swimming pool environments

1.2 Literature Review

Recent technological developments in monitoring systems have introduced advanced solutions for improving safety and management in public facilities. In the context of swimming pools, traditional supervision methods mainly rely on lifeguards and manual observation to ensure the safety of swimmers. While human supervision is essential, it can sometimes be challenging to maintain constant attention in crowded environments. As a result, researchers have explored the use of intelligent technologies to support automated monitoring and reduce the risk of accidents such as drowning.

One of the key technologies applied in this area is **computer vision**, which enables machines to interpret and analyze visual information captured through cameras. Computer vision techniques can process video data to detect objects, recognize human activities, and analyze movement patterns. In swimming pool environments, camera-based systems can be used to observe swimmers continuously and analyze their behavior in real time. By studying body posture, motion patterns, and activity duration, these systems can identify unusual behaviors that may indicate distress or potential drowning situations.

Several studies have applied **machine learning and deep learning algorithms** to improve the accuracy of behavior recognition in surveillance systems. Models such as convolutional neural networks (CNNs) and object detection frameworks have been widely used to analyze human movement and detect abnormal activities in video streams. These techniques allow automated systems to distinguish between normal swimming behavior and irregular movement patterns that could signal an emergency. The use of intelligent algorithms therefore enhances the capability of monitoring systems to identify dangerous situations more quickly.

Another research area related to pool safety involves **activity recognition and behavioral analysis in aquatic environments**. By analyzing motion trajectories and swimmer positions within video frames, computer vision models can track swimmer movements and evaluate their activity patterns. Such information can help identify behaviors such as prolonged inactivity, struggling movements, or irregular swimming patterns that may require immediate attention from lifeguards or pool operators.

Despite the progress made in computer vision–based monitoring systems, many existing solutions are primarily designed for **general surveillance rather than specialized drowning detection**. In many cases, video monitoring systems simply record footage without performing intelligent analysis of swimmer behavior. Furthermore, some systems lack real-time alert mechanisms that could assist lifeguards in responding quickly to emergencies.
Although research has demonstrated the effectiveness of computer vision and artificial intelligence in activity recognition and safety monitoring, their application in **swimming pool drowning detection systems** is still developing. Therefore, there is a need for an intelligent monitoring solution that can automatically analyze swimmer behavior and detect potential drowning situations in real time. Integrating such computer vision techniques within smart pool management systems can significantly enhance swimmer safety and support faster emergency response.

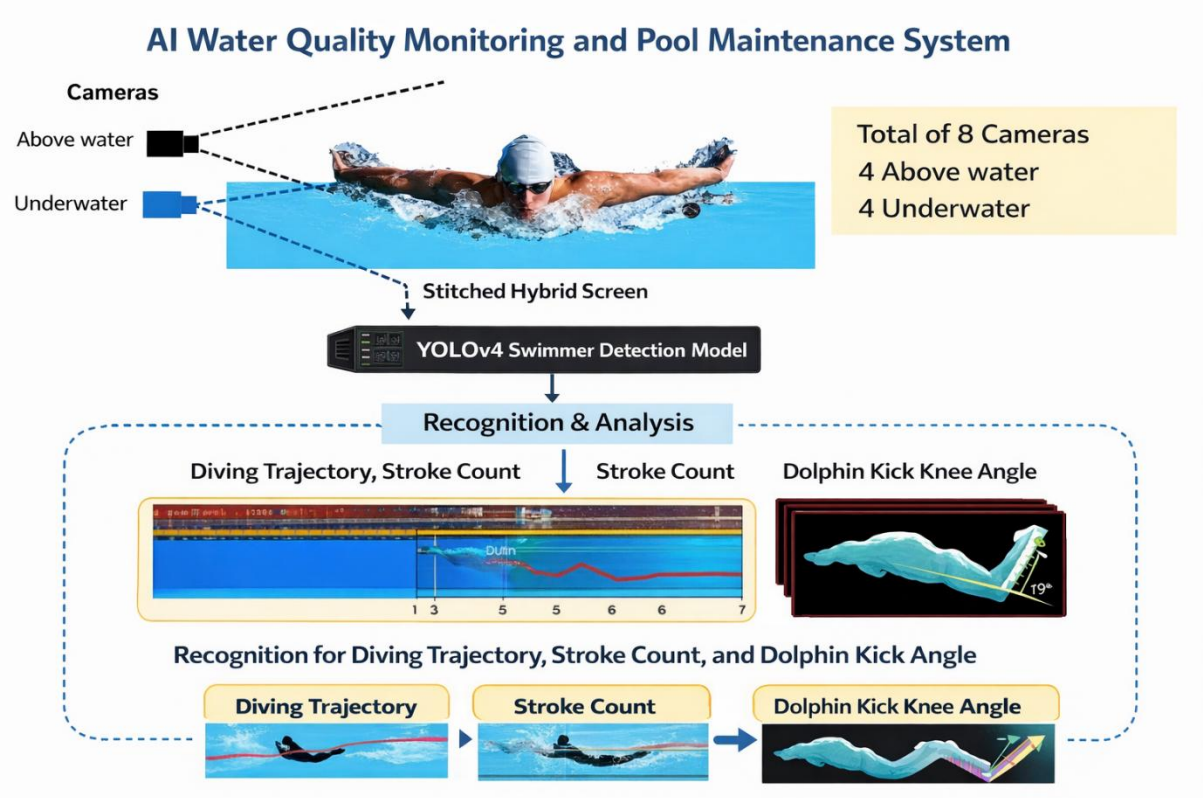


Figure 1.1 Swimmer Detection Process

1.3 Research Gap

Although various studies have investigated the use of surveillance cameras and monitoring systems in swimming pool environments, many existing solutions mainly serve as **basic video recording or general security monitoring tools**. These systems capture video footage of the pool area, but they typically lack intelligent mechanisms to analyze swimmer activities or automatically detect dangerous situations. As a result, identifying swimmers who may be experiencing distress still depends heavily on manual observation by lifeguards.

In most swimming pool facilities, the responsibility for detecting potential drowning incidents relies entirely on lifeguards who continuously observe the pool area. While lifeguards are trained to respond to emergencies, maintaining constant attention for long periods can be challenging, especially in crowded environments where multiple swimmers are present simultaneously. This limitation increases the possibility that abnormal swimmer behaviors, such as struggling or prolonged inactivity, may go unnoticed for critical moments.

Furthermore, many existing monitoring systems do not incorporate **advanced computer vision or artificial intelligence techniques** to analyze swimmer behavior in real time. Without intelligent behavior analysis, these systems are unable to distinguish between normal swimming activities and movements that may indicate distress or potential drowning situations. The absence of automated detection mechanisms limits the ability of such systems to provide early warnings or assist lifeguards in responding quickly to emergencies.

Therefore, a clear research gap exists in the development of **intelligent swimmer monitoring systems that can automatically analyze behavioral patterns and detect potential drowning situations in real time**. The proposed **Computer Vision-Based Swimmer Behavior and Drowning Detection Module** aims to address this gap by applying machine learning and video analysis techniques to monitor swimmer activities and identify abnormal movement patterns. By introducing automated behavior analysis and real-time alert mechanisms, the proposed approach can enhance swimmer safety, support lifeguards in emergency detection, and contribute to safer swimming pool environments.

2. OBJECTIVE

2.1 Main Objective

To design and develop an **AI-powered computer vision–based swimmer behavior monitoring and drowning detection module** that continuously analyzes video data captured from swimming pool environments in order to automatically identify swimmer activities and detect abnormal behavior patterns that may indicate distress or potential drowning situations. The proposed system aims to utilize advanced **computer vision and machine learning techniques** to recognize movement patterns, body posture, and activity duration of swimmers in real time. By accurately distinguishing between normal swimming behavior and potentially dangerous situations, the system will generate **timely alerts and notifications** for lifeguards or pool operators. This intelligent monitoring approach is intended to support faster emergency response, enhance swimmer safety, and contribute to the overall effectiveness of the **AI-based Water Quality Monitoring and Pool Maintenance System** by integrating automated safety monitoring within the pool management environment.

2.2 Specific Objectives

- To continuously capture and process video streams from swimming pool surveillance cameras for detailed monitoring of swimmer movements and behavior patterns.

- To identify and analyze irregular or abnormal swimmer behaviors, such as prolonged inactivity, struggling motions, or unusual postures, which could indicate distress or potential drowning incidents.

- To develop and train a robust machine learning and computer vision framework capable of distinguishing between normal swimming activities and high-risk behaviors with high accuracy.

- To implement a real-time alerting system that automatically notifies lifeguards or pool supervisors whenever potentially dangerous swimmer behavior is detected, facilitating rapid intervention.
- To integrate temporal and spatial analysis of swimmer behavior, enabling the system to track individual swimmers over time and provide early detection of cumulative risks.

- To evaluate the system's effectiveness in enhancing swimmer safety by conducting simulations or controlled testing in pool environments, measuring detection accuracy, response times, and false alarm rates.

- To investigate how the computer vision module can complement the broader AI-based Water Quality Monitoring and Pool Maintenance System by contributing to overall pool safety management.

- To provide recommendations for system deployment, usability, and operational guidelines for pool facilities to maximize safety and ensure reliable real-time monitoring.

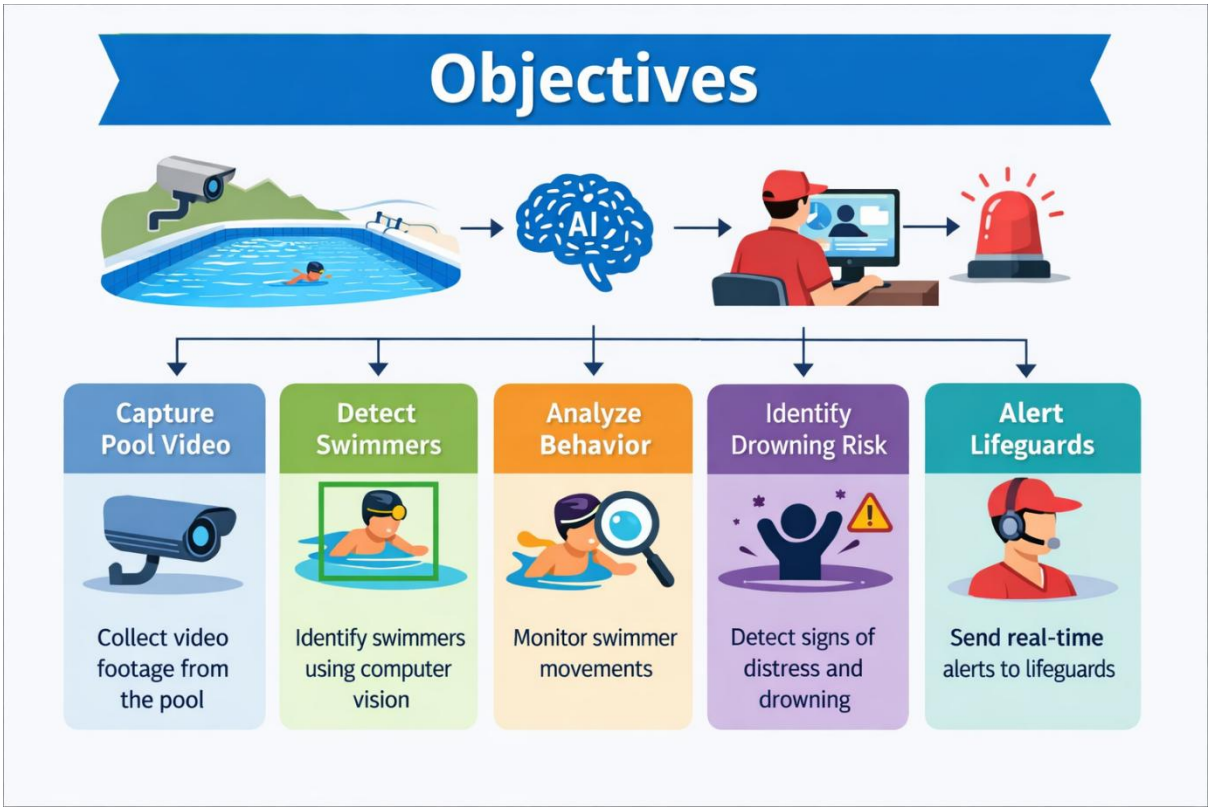


Figure 2.1 AI Drowning Detective System

3. METHODOLOGY

The proposed research adopts a **Design Science Research (DSR) approach**, which emphasizes the creation and evaluation of innovative solutions to solve real-world problems. In this study, the real-world problem addressed is the challenge of ensuring swimmer safety in swimming pools, where traditional monitoring relies heavily on human observation and is prone to delays or oversight in detecting drowning or distress situations. The research aims to develop a **computer vision–based swimmer behavior and drowning detection system** that can automatically analyze swimmer activities in real time and provide immediate alerts for potential emergency situations.

The research methodology is structured around the **design, implementation, and evaluation** of an intelligent monitoring system. The first step involves collecting video data from swimming pool environments. This data can come from either **live camera feeds** or **simulated datasets** representing different swimmer behaviors under various conditions. Ethical considerations are a key component of the data collection process to ensure that privacy and confidentiality are maintained, particularly when using video footage of individuals.

Once the video data is collected, it will be preprocessed to improve the accuracy of the analysis. Preprocessing steps include **frame extraction, background subtraction, noise reduction, and normalization of lighting conditions**. These steps ensure that the subsequent analysis is robust and reliable. The preprocessed video frames are then fed into **computer vision algorithms** that detect swimmers and track their movements over time. Object detection and

tracking frameworks such as **YOLO (You Only Look Once)** or **OpenCV-based detection pipelines** will be used to localize and follow swimmer positions in real time.

The next stage involves **behavior analysis using machine learning and deep learning techniques**. By training convolutional neural networks (CNNs) or other suitable architectures on labelled datasets of swimmer actions, the system can classify different types of behavior, distinguishing normal swimming from potentially dangerous actions, such as prolonged inactivity, erratic movement, or struggling motions. The system will also incorporate **temporal analysis**, considering movement patterns over time to improve the detection of high-risk situations that may develop gradually.

A key component of the methodology is the **real-time alert mechanism**. When the system identifies swimmer behavior that meets predefined criteria for distress, it will automatically generate alerts to notify lifeguards or pool supervisors. The alerts can be visual, auditory, or integrated into a management dashboard for immediate intervention. This automated alerting approach ensures faster response times compared to traditional human observation alone.

The system will be developed using **Python** as the primary programming language due to its extensive libraries for computer vision (**OpenCV, TensorFlow, PyTorch**) and real-time data processing. A **MySQL or similar database** will be used to store video metadata, swimmer behavior records, and system alerts, enabling retrospective analysis and model improvement. Additional software tools for visualization and monitoring dashboards may be employed to provide lifeguards with a clear, real-time overview of pool safety.

To evaluate the performance of the proposed system, extensive testing will be conducted using both **simulated datasets** and real video data from swimming pools. Key performance metrics will include **detection accuracy, false positive and false negative rates, system response time, and the effectiveness of alerts in prompting timely intervention**. Comparative analysis will also be performed to demonstrate how the proposed automated system improves safety and response efficiency compared to conventional manual monitoring practices.

Finally, the methodology will incorporate an **iterative design and testing cycle**, where insights from initial tests are used to refine the behavior detection models, improve alert algorithms, and enhance the system's robustness under varying environmental conditions, such as lighting changes, swimmer density, and camera angles. This iterative approach ensures that the final system is reliable, scalable, and capable of supporting real-world deployment in swimming pools, significantly enhancing swimmer safety and complementing the broader AI-based Water Quality Monitoring and Pool Maintenance System.

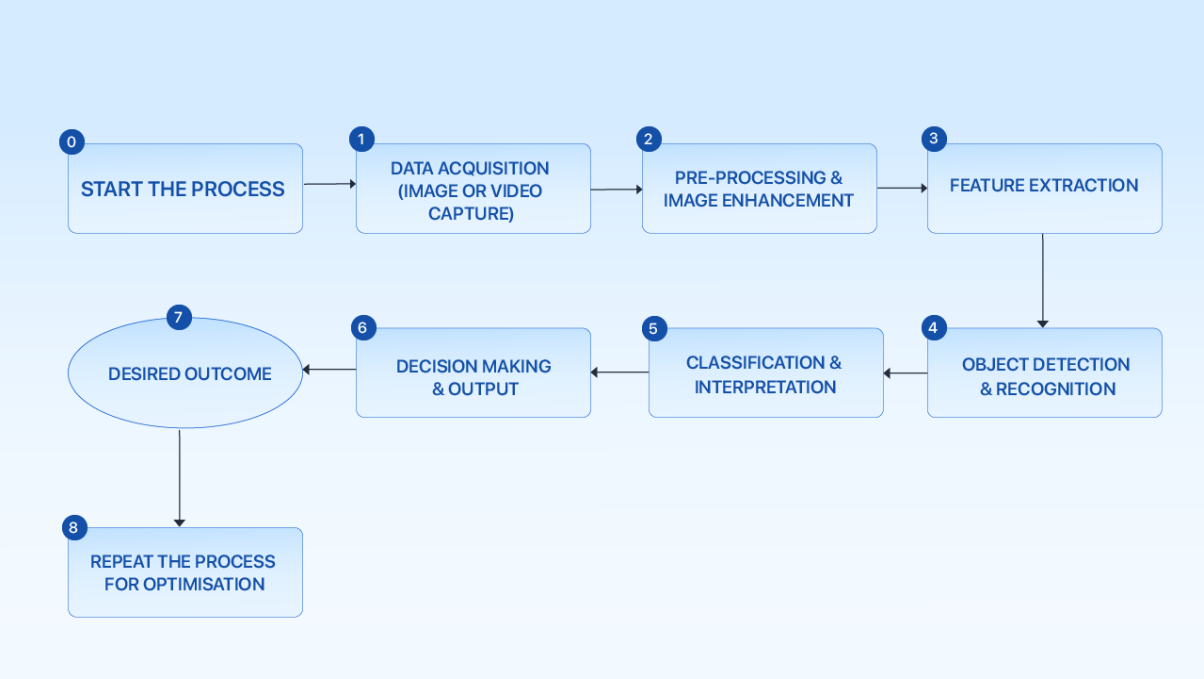


Figure 3.1 Computer Vision Processing Pipeline

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed **Computer Vision–based Swimmer Behavior and Drowning Detection System** is designed as an intelligent safety monitoring platform that integrates seamlessly into the broader **AI-based Water Quality Monitoring and Pool Maintenance System**. The system is composed of multiple interrelated components that collectively enhance swimmer safety, provide real-time alerts, and support pool management operations. Core modules of the system include **video data acquisition, swimmer detection and tracking, behavior analysis, real-time alerting, and administrative monitoring dashboards**. These modules work together to continuously capture, process, and interpret swimmer activity to identify potential drowning or distress situations.

At the foundation of the system is the **video acquisition layer**, where surveillance cameras capture continuous live footage of the pool environment. The video streams are preprocessed to improve quality and reduce noise, incorporating operations such as **frame extraction, background subtraction, lighting normalization, and object segmentation**. Preprocessing ensures that the subsequent analysis can accurately detect and track swimmers in different pool conditions, including varying lighting, water reflections, and swimmer density.

The next layer is the **swimmer detection and tracking module**, which employs **computer vision algorithms** such as object detection models (e.g., YOLO) and motion tracking techniques to identify and monitor individual swimmers in real time. This module maintains temporal tracking of swimmer positions and movement trajectories, enabling the system to observe patterns and detect changes in behavior over time.

The **behavior analysis module** is the core of the system. Using **deep learning models**, such as convolutional neural networks (CNNs), the system classifies swimmer actions and identifies abnormal behavior patterns that may indicate distress, such as irregular motion, prolonged inactivity, or struggling movements. This module also incorporates **temporal and spatial analysis**, allowing the system to evaluate behavior over time and across multiple regions of the pool, which enhances early detection of potential drowning incidents.

The **alert management module** is responsible for generating **real-time notifications** when high-risk behaviors are detected. Alerts can be delivered via **visual dashboards, audio alarms, or direct notifications to lifeguards and pool operators**, ensuring rapid response to emergencies. This automated alerting reduces reliance solely on human observation and improves the chances of timely intervention.

All processed data, including swimmer behavior logs, alerts, and video metadata, is stored in a **centralized, secure database**, allowing historical analysis, model retraining, and continuous system improvement. Administrative users can access a **dashboard interface** to monitor pool safety, review detected incidents, and analyze swimmer behavior trends over time. This interface supports visualization of swimmer locations, detected risks, and system performance metrics.

The architecture is designed to support **scalability**, enabling the integration of additional cameras, AI models, or pool monitoring components in the future. Security and privacy

considerations are incorporated throughout, including **controlled access to video data, secure storage, and adherence to ethical guidelines for data use**.

By integrating these components, the proposed system provides a **comprehensive, intelligent safety monitoring solution** for swimming pools. It enhances lifeguard capabilities, improves emergency response times, and contributes to safer swimming environments while complementing the broader water quality and pool maintenance management framework.

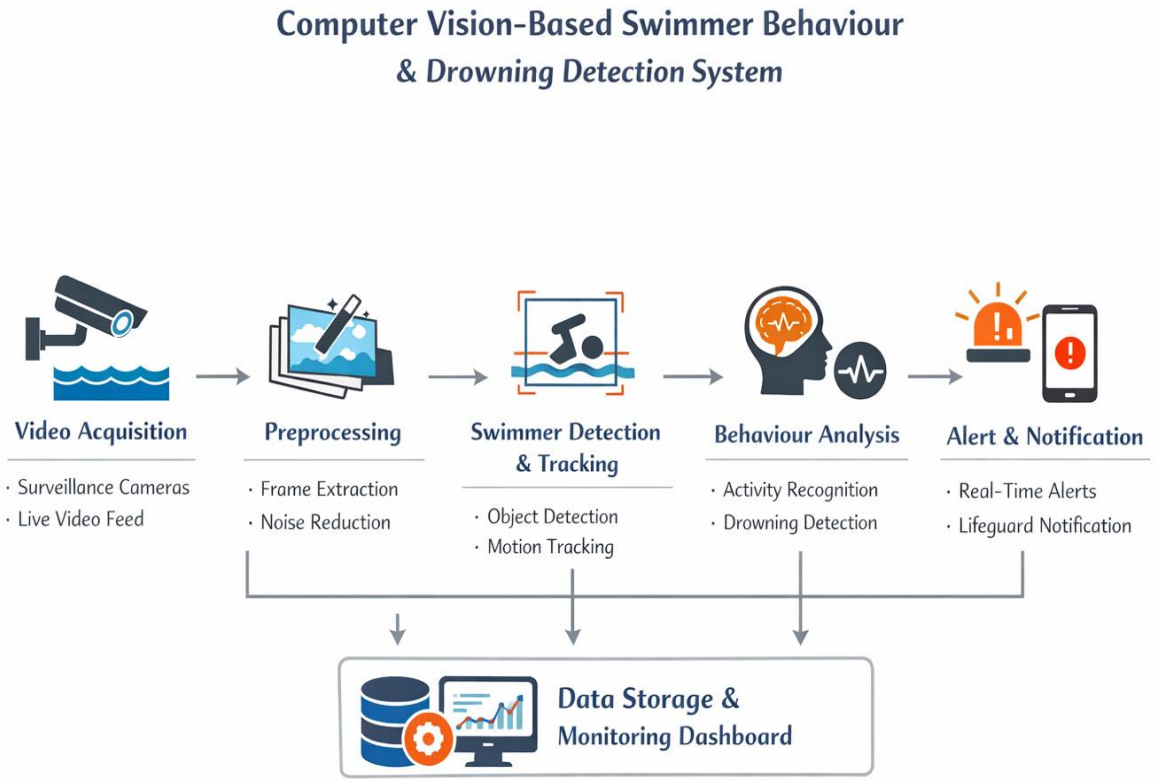


Figure 4.1 Computer Vision-Based Swimmer Behaviour and Drowning Detection System

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed **Computer Vision–based Swimmer Behavior and Drowning Detection System** will serve multiple stakeholders involved in swimming pool safety and management operations:

- **Pool Managers** – They are responsible for overseeing the overall safety of pool operations. Using the system's monitoring dashboard, pool managers can observe real-time swimmer activity, receive automated alerts for potential distress situations, and access historical behavior analytics to support safety planning and incident prevention.
- **Lifeguards/ Pool Safety Staff** - These stakeholders are on the front line of swimmer safety. The system will provide them with **real-time notifications and visual cues** when abnormal or high-risk swimmer behavior is detected, enabling faster response and intervention during emergencies.
- **Maintenance Staff** – While not directly involved in swimmer safety detection, maintenance staff may benefit indirectly from insights provided by the system regarding swimmer density and activity patterns, which can influence water quality and maintenance planning.
- **Facility Administrators** – Responsible for ensuring the facility complies with safety regulations and operational standards, administrators can use the system's dashboards to generate reports on safety incidents, review alert logs, and evaluate the effectiveness of the safety monitoring system over time.
- **Swimmers / Pool Users** – Although they do not directly operate the system, swimmers are the primary beneficiaries. The system enhances safety by reducing the risk of drowning incidents and ensuring faster emergency response, creating a safer and more secure swimming environment.
- **Regulatory Authorities/ Safety Inspectors** - These stakeholders may use aggregated reports from the system to verify compliance with local safety standards, evaluate safety protocols, and identify areas for improvement in swimmer safety management.

- **System Administrators/ IT support** - Responsible for the configuration, maintenance, and security of the system, including camera setup, database management, and ensuring continuous availability of real-time monitoring and alerts.

5.2 Functional Requirements

The proposed module is designed to enhance swimmer safety by automatically monitoring behaviors and detecting potential drowning incidents. The main functional requirements of the system are as follows:

1. **Video Data Acquisition:** The system shall continuously capture live video streams from surveillance cameras installed in the swimming pool environment.
2. **Preprocessing of Video Data:** The system shall preprocess video frames to improve detection accuracy, including operations such as background subtraction, noise reduction, and normalization of lighting conditions.
3. **Swimmer Detection:** The system shall detect and localize individual swimmers within each video frame using computer vision techniques.
4. **Swimmer Tracking:** The system shall track the movement of swimmers over time, maintaining a unique identifier for each swimmer to analyze motion trajectories.
5. **Behavior Analysis:** The system shall analyze swimmer movements and activity patterns to identify abnormal behaviors that may indicate distress, such as prolonged inactivity, struggling, or irregular motion.
6. **Drowning Detection:** The system shall determine potential drowning or high-risk situations based on predefined behavioral criteria and real-time analysis.
7. **Real-Time Alerting:** The system shall automatically generate alerts for lifeguards or pool operators when abnormal swimmer behavior or potential drowning is detected. Alerts may be delivered via visual indicators, audio alarms, or notifications on a dashboard interface.
8. **Historical Data Logging:** The system shall store swimmer behavior records, detected incidents, and alert history in a secure database for later analysis and performance evaluation.

9. **Dashboard Visualization:** The system shall provide a user-friendly interface that displays swimmer locations, real-time alerts, incident logs, and behavioral analytics for pool managers and lifeguards.

5.3 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance standards, and operational constraints of the system. For the proposed **Computer Vision–based Swimmer Behavior and Drowning Detection System**, the key non-functional requirements are as follows:

- **Performance:**
  The system shall process video streams and detect abnormal swimmer behavior in real time, generating alerts promptly to enable timely intervention by lifeguards or pool staff. Detection latency should be minimal to ensure swimmer safety.
- **Security:**
  Video footage, alert logs, and behavioral data shall be stored securely, with access restricted to authorized personnel. The system shall comply with ethical and privacy guidelines to protect swimmer identities and sensitive information
- **Usability:**
  The user interface and dashboard shall be intuitive and easy to navigate, allowing lifeguards, pool managers, and administrators to quickly interpret alerts, review swimmer activity, and respond to emergencies efficiently.
- **Scalability:**
  The system shall be capable of handling multiple cameras and large pools with high swimmer density, and it should support deployment across multiple facilities if needed without compromising performance.
- **Reliability:**
  The system shall operate continuously during pool hours, providing uninterrupted monitoring and alerting services. It should be robust against software crashes, camera failures, or network interruptions.

- **Accuracy:**

  The behavior recognition and drowning detection algorithms shall maintain a high level of accuracy, minimizing false positives and false negatives to ensure reliable identification of potential distress situations.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed **Computer Vision–based Swimmer Behavior and Drowning Detection System** is designed for organizations and facilities that manage swimming pools where swimmer safety is a critical concern. The primary target market includes **hotels, sports complexes, public swimming pools, fitness centers, educational institutions with pool facilities, and recreational water parks**. These organizations require advanced safety monitoring solutions to ensure real-time supervision of swimmers and rapid response to potential emergency situations.

The main users of the system are **pool managers, lifeguards, and facility supervisors** who are responsible for maintaining a safe swimming environment. These users need a **reliable and intelligent system** that can continuously monitor swimmer behavior, identify abnormal or high-risk actions, and provide immediate alerts for timely intervention.

Additionally, **facility administrators** are an important user group, as they require comprehensive reporting and historical data on pool safety incidents, swimmer activity trends, and system performance. The system provides them with actionable insights to improve safety policies, optimize staffing, and comply with regulatory standards.

Although **swimmers and pool users** do not directly operate the system, they are a key stakeholder group, benefiting from improved safety measures, quicker emergency response, and reduced risk of accidents.

The system is particularly suited for **high-traffic or crowded pool environments**, where human supervision alone may be insufficient to detect all potential drowning incidents. By integrating automated behavior analysis, real-time alerts, and continuous monitoring, the system meets the needs of facilities that prioritize **swimmer safety, operational efficiency, and compliance with safety regulations**.

6.2 Value Proposition

The proposed system delivers an **intelligent, automated solution for enhancing swimmer safety and emergency response in swimming pools**. Unlike traditional safety monitoring methods that rely solely on human observation, the system leverages **real-time computer vision and AI-based behavior analysis** to detect abnormal or high-risk swimmer actions automatically.

By continuously tracking swimmer movements and identifying patterns indicative of distress or potential drowning, the system enables **lifeguards and pool managers to respond faster and more effectively**. This reduces the likelihood of accidents and improves overall pool safety.

The module adds value by providing **real-time alerts, behavioral analytics, and historical incident data**, allowing facility operators to make informed decisions about staffing, monitoring priorities, and safety protocols. High-risk scenarios are flagged immediately, ensuring that intervention is timely, precise, and efficient.

Moreover, the system is designed to operate seamlessly in **high-traffic or crowded pool environments**, where human supervision alone may be insufficient. By combining automated monitoring with actionable insights, the system enhances operational efficiency, supports regulatory compliance, and provides **peace of mind to both facility operators and swimmers**.

6.3 Revenue Model

The proposed system can be commercialized as an **intelligent safety monitoring platform** for swimming pools. Facility operators and pool management organizations can adopt the system to gain **real-time insights into swimmer behavior, automated drowning detection, and emergency alerting capabilities**.

Potential revenue models include:

- **Subscription-Based Model:** Facilities can pay **monthly or annual subscription fees** to access the system's cloud-based monitoring dashboards, real-time alerts, and behavioral analytics tools.
- **On-Premise Software Licensing:** Operators can purchase a **one-time license** to deploy the system locally within their facility, including access to the behavior detection module and video analytics features.

- **Safety Monitoring Service Packages:** Offering **value-added services**, such as incident reporting, behavioral analysis summaries, and system performance evaluations, to assist facilities in improving swimmer safety and regulatory compliance.
- **Integration Consulting and Customization Services:** Providing professional support for **custom camera setups, AI model tuning, and dashboard personalization**, which ensures the system is optimized for unique pool environments and operational requirements.

This revenue model allows the system to generate income while offering **flexible options for facilities of varying sizes and budgets**, ensuring wider adoption and enhanced swimmer safety across different types of pool environments

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6.4 Cost Estimation

The estimated cost of developing and deploying the **Computer Vision–based Swimmer Behavior and Drowning Detection System** includes expenses associated with software development, AI model training, camera and hardware installation, and secure data storage. Additional costs may arise from setting up the monitoring infrastructure, maintaining the system, and integrating it with existing pool management platforms.

Key cost components include:

- **Software Development:** Designing and implementing the behavior detection algorithms, real-time alerting system, and user dashboards.
- **Hardware and Cameras:** Purchasing and installing high-resolution surveillance cameras, networking equipment, and optional edge computing devices for local processing.
- **AI Model Training and Maintenance:** Preparing and labelling video datasets, training deep learning models, and periodically updating them to improve accuracy.
- **Data Storage and Cloud Services:** Storing video metadata, behavioral logs, and system alerts securely, either on-premises or through cloud infrastructure.
- **System Maintenance and Support:** Ongoing software updates, troubleshooting, and periodic system calibration to ensure reliable operation.
- **Integration and Customization:** Optional costs for adapting the system to different pool layouts, camera angles, or specific safety protocols.

While the initial investment may be considerable, the system is expected to **reduce long-term operational risks and costs** by improving swimmer safety, minimizing emergency response delays, and enhancing regulatory compliance. By preventing accidents and ensuring timely intervention, the system can also mitigate potential liability costs for facility operators, providing both **financial and safety benefits** over time.

6.5 Pricing Strategy

The pricing of the **Computer Vision–based Swimmer Behavior and Drowning Detection System** can be structured according to the size of the facility, the number of cameras deployed, and the level of AI-driven monitoring required. Facilities with smaller pools or lower swimmer traffic may opt for a **basic monitoring package**, which includes core behavior detection and real-time alerting features.

Larger facilities, such as hotels, sports complexes, or high-traffic public pools, may subscribe to **advanced service tiers** that offer additional benefits, including detailed behavioral analytics, historical incident reporting, multi-camera integration, and customization options for different pool layouts.

A **tiered pricing model** can be implemented, providing flexibility for facilities to choose a plan that matches their operational requirements and budget. Optional add-ons, such as **custom AI model training, extended dashboard features, or integration with other pool management systems**, can be offered at an additional cost to further enhance safety monitoring capabilities.

This approach ensures that the system is **affordable for small to medium-sized facilities** while providing **scalable solutions for larger or more complex pools**, making it attractive across a wide range of potential users. The tiered structure also encourages adoption by allowing facilities to start with essential features and upgrade as their needs grow

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6.6 Competitive Advantage

The proposed **Computer Vision–based Swimmer Behavior and Drowning Detection System** offers significant advantages over traditional pool safety monitoring methods, which largely rely on human observation and manual supervision. By leveraging **real-time AI-driven behavior analysis**, the system can automatically detect abnormal swimmer actions, flag potential drowning incidents, and provide immediate alerts to lifeguards or pool managers.

Key competitive advantages include:

- **Automated Safety Monitoring:** Unlike conventional methods, the system continuously monitors all swimmers, reducing the likelihood of missed distress signals and human error.
- **Real-Time Alerts and Intervention:** The system generates instant notifications when high-risk behaviors are detected, enabling faster response times and minimizing the risk of accidents.
- **Behavioral Analytics:** The system records swimmer activity and tracks movement patterns over time, providing actionable insights for improving pool safety protocols and optimizing lifeguard deployment.
- **Scalability and Adaptability:** It can be deployed in facilities of various sizes, from small community pools to large public or commercial swimming complexes, and can integrate with multiple camera feeds for comprehensive coverage.

- **Enhanced Compliance and Reporting:** By maintaining detailed incident logs and behavioral data, the system supports regulatory compliance and provides evidence-based reports for facility management and safety audits.
- **Reduced Liability and Operational Risk:** By improving emergency response and monitoring coverage, the system helps reduce liability exposure for facility operators while enhancing the overall safety of pool users.

By combining **computer vision, AI-driven detection, and real-time alerts**, the system provides a **proactive and intelligent solution** that goes beyond traditional supervision, making it a safer, more reliable, and technologically advanced option for modern swimming pool facilities.

6.7 Intellectual Property Considerations

The proposed **Computer Vision–based Swimmer Behavior and Drowning Detection System** incorporates novel AI algorithms, real-time behavior analysis techniques, and system designs for automated drowning detection. These innovations represent **intellectual assets** that can be protected through software copyrights, patent filings, or other forms of intellectual property registration.

Securing intellectual property rights for the system will ensure that the **unique AI models, detection algorithms, and safety monitoring processes** remain proprietary, allowing the developers to retain ownership and control over the technology.

Intellectual property protection also supports the **commercial deployment of the system**, enabling licensing opportunities, partnerships with pool management companies, and future enhancements without the risk of unauthorized replication. Additionally, it provides a foundation for **further research and development**, encouraging the integration of advanced AI models, expanded safety features, and multi-facility scalability in future versions.

By safeguarding the system's intellectual property, the developers can **maximize both the commercial potential and technological value** of the solution, while contributing to innovation in smart safety monitoring for swimming pools

7. BUDGET AND JUSTIFICATION

The estimated budget for developing the **Computer Vision–Based Swimmer Behaviour and Drowning Detection Module** mainly includes expenses related to camera equipment for video capture, cloud storage for video data processing, and development resources required for building the AI detection model. Since the project is implemented as a **research prototype**,

most of the software tools and machine learning frameworks used in development are open-source and freely available.

Item	Estimated Cost (LKR)	Justification
Camera Equipment / Video Capture Devices	30,000	Basic surveillance cameras or webcams will be used to capture swimming pool video footage required for swimmer detection and behavior analysis.
Software Tools and Libraries	0	Development tools such as Python, OpenCV, TensorFlow/PyTorch, and other machine learning libraries are open-source and freely available.
Cloud Storage and Computing Services	7,000	Cloud services may be required to store video datasets and support model testing, processing, and system deployment.
Development and Research Effort	20,000	Represents the estimated effort required for system design, AI model development, testing, and implementation during the research project
Miscellaneous Expenses	5,000	Covers internet usage, documentation preparation, printing, and other small development-related expenses.

Table 7.1 Budget and Justification

Total Estimated Budget: LKR 62,000

Budget Justification

The proposed budget is designed for a **research prototype rather than a full commercial system**. Open-source development tools significantly reduce software costs, while basic sensors are sufficient for testing purposes. Cloud infrastructure requirements are minimal since

the system will mainly be used for experimental evaluation during the research project. Therefore, the overall cost of developing the prototype system remains relatively low while still supporting the necessary research activities.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure outlines the major activities required for developing the **Computer Vision–Based Swimmer Behavior and Drowning Detection Module** as part of the **Smart Swimming Pool Safety and Monitoring System**. The project tasks are divided into several development phases to ensure effective time management, systematic progress, and coordinated collaboration among all project members. Since the project is implemented as a **group-based research system**, different modules are developed simultaneously by team members while maintaining integration with the overall system.

The first stage of the project focuses on **problem investigation and requirement identification**. During this phase, the team reviews existing research related to swimming pool safety systems, computer vision–based monitoring, and drowning detection technologies. Academic publications, technical documentation, and existing safety monitoring solutions are examined to understand current limitations and identify opportunities for improvement. At the same time, the functional and technical requirements of each module are defined so that all components can be developed concurrently.

Following the requirement analysis, the project moves into the **system design and architecture planning phase**. In this stage, the overall architecture of the smart monitoring system is established, identifying how different modules—such as water quality monitoring, swimmer detection, behavior analysis, and alert management—interact with one another. For the drowning detection component, the design includes defining the video processing workflow, data input sources, behavior analysis logic, and alert generation mechanisms. The communication between the computer vision module and other system components is also structured during this phase.

After the design is finalized, the **development and implementation stage** begins. Each team member works on their assigned module simultaneously while ensuring compatibility with the shared system architecture. For the drowning detection module, development activities include implementing computer vision algorithms for swimmer detection, analyzing swimmer movement patterns, and identifying abnormal behaviors that may indicate distress or potential drowning situations. Machine learning and image processing techniques are integrated to enable the system to process video data and detect risky situations in real time.

The next phase focuses on **system integration and functional testing**. In this stage, all modules developed by the project members are connected to create a unified monitoring system. The drowning detection module is tested using recorded or simulated video datasets to evaluate its ability to recognize swimmer movements and detect dangerous situations

accurately. Integration testing ensures that the module can communicate effectively with other components such as the monitoring dashboard and alert system.

The final stage of the project involves **performance evaluation, refinement, and documentation**. The effectiveness of the detection algorithms is analyzed using evaluation metrics such as detection accuracy, response time, and reliability of alerts. Any identified limitations are addressed through further improvements to the algorithms or system configuration. The development process, experimental results, and system architecture are documented as part of the final research report.

By distributing responsibilities across multiple modules and allowing parallel development among team members, the project ensures efficient progress and balanced workload management. The **Swimmer Behavior and Drowning Detection Module** plays a critical role within the overall system by enhancing swimmer safety through intelligent real-time monitoring and automated risk detection.

8.1. Task Breakdown per member

Member ID	Component		Key Tasks
IT22156310	Crowd-Aware Maintenance Scheduling Module		Analyze swimmer crowd density using CCTV data, estimate crowd density level (CDL), integrate sensor data, generate dynamic maintenance recommendations.
IT22247254	Water Quality Classification and Prediction Module		Collect water quality sensor data, preprocess data, develop machine learning models to classify water quality conditions, analyze importance of water quality parameters.
IT22248480	Swimmer Behavior and Drowning Detection Module		Process video streams from CCTV cameras, detect abnormal swimmer behaviors, identify drowning risk situations, generate emergency alerts.
IT22277336	Garbage Intrusion Detection Module		Detect garbage objects entering the pool, analyze human actions related to garbage disposal, trigger alerts for pool management.

Table 8.1 Task Breakdown per Member

9 . GANTT CHART

- Insert Gantt chart here (Semester-wise breakdown with milestones).

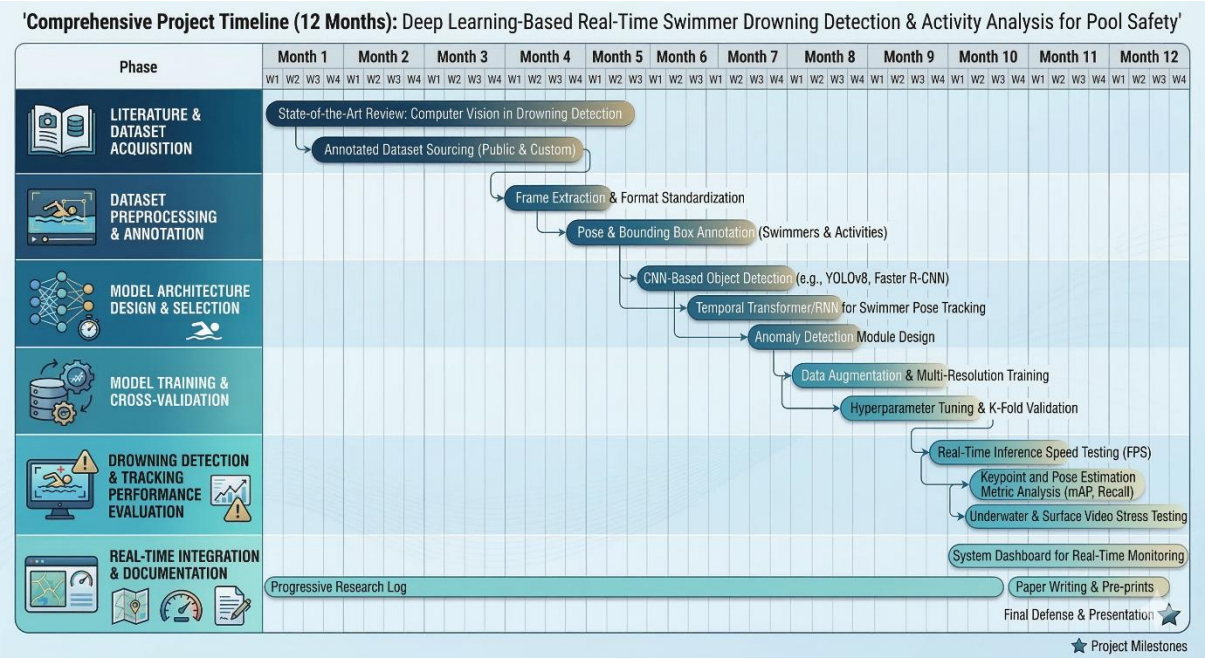


Figure 9.1 Gantt Chart

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9. REFERENCES LIST

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10. APPENDICES

- Appendix A – Computer Vision Processing Workflow
- Appendix B – Dataset Description
- Appendix C – Algorithm Overview
- Appendix D – Software Tools and Technologies
- Appendix E – Sample System Interface